# A Triangle with a Two-Leaf Broom: A Rooted Supporting-Plane Inequality

**Abstract**

Let $B$ be obtained from a triangle by joining one triangle vertex to the centre of a two-leaf star. We prove that every graphon $W$ of edge density $p \geq 1/2$ satisfies

$$t(B, W) \geq p^4(2p - 1) = \Phi_B(p).$$

The proof is edge-density-sensitive. It does not use the false universal comparison between the associated five-edge tree and $S_4 \sqcup 2K_2$. Instead, two pointwise graphon inequalities reduce the problem to a sharp two-variable scalar inequality on

$$0 \leq a \leq d, \qquad a \geq d - (1 - p).$$

The scalar lower bound contains a multiple of $a - d^2$, whose integral is zero for $a = T_W d$. This proves the formerly open NetworkX Atlas graph 100.

## 1 The graph, its target, and rooted quantities

Let $(\Omega, \mu)$ be an arbitrary probability space and let $W \colon \Omega^2 \to [0, 1]$ be symmetric and measurable. Put

$$p = \int_{\Omega^2} W(x, y) \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y), \qquad d(x) = \int_{\Omega} W(x, y) \, \mathrm{d}\mu(y),$$

and define

$$A = T_W d, \qquad C = T_W(d^2).$$

The rooted triangle density is

$$\tau(x) = \int_{\Omega^2} W(x, y) W(x, z) W(y, z) \, \mathrm{d}\mu(y) \, \mathrm{d}\mu(z).$$

Construct $B$ from a triangle rooted at $o$ by adding three new vertices $u, v, w$ and exactly the three further edges

$$ou, \qquad uv, \qquad uw.$$

After fixing the image $x$ of $o$, integrating $v, w$, and then $u$ gives the exact identity

$$t(B, W) = \int_{\Omega} \tau(x) C(x) \, \mathrm{d}\mu(x). \tag{1}$$

The graph has six vertices and six edges. After properly colouring the triangle, each successive vertex of the attached tree has $x - 1$ choices. Therefore

$$\chi_B(x) = x(x - 1)^4(x - 2), \qquad \chi(B) = 3.$$

For $q = 1 - p > 0$,

$$q^6 \chi_B(1/q) = p^4(2p - 1).$$

Thus the polynomial continuation of the target is

$$\Phi_B(p) = p^4(2p - 1), \tag{2}$$

and the complete required interval is $p \in [1/2, 1]$.

## 2 Two pointwise graphon reductions

**Lemma 2.1** (Rooted feasible region)**.** *For almost every $x$, writing $d = d(x)$ and $a = A(x)$, one has*

$$0 \le a \le d, \qquad a \ge d - (1 - p).$$

*Moreover,*

$$\int_\Omega A \, \mathrm{d}\mu = \int_\Omega d^2 \, \mathrm{d}\mu.$$

*Proof.* Nonnegativity is immediate. Since $d(y) \le 1$ almost everywhere,

$$A(x) = \int W(x, y) d(y) \, \mathrm{d}\mu(y) \le d(x).$$

The elementary inequality $rs \ge r + s - 1$ for $r, s \in [0, 1]$, applied to $r = W(x, y)$ and $s = d(y)$, gives

$$A(x) \ge \int \big(W(x, y) + d(y) - 1\big) \, \mathrm{d}\mu(y) = d(x) + p - 1.$$

Finally Tonelli's theorem and symmetry give

$$\int A(x) \, \mathrm{d}\mu(x) = \int_{\Omega^2} W(x, y) d(y) \, \mathrm{d}\mu^2 = \int_\Omega d(y)^2 \, \mathrm{d}\mu(y).$$

□

**Lemma 2.2** (Broom and triangle roots)**.** *For almost every $x$,*

$$C(x) \ge \begin{cases} A(x)^2/d(x), & d(x) > 0, \\ 0, & d(x) = 0, \end{cases} \qquad \tau(x) \ge \big(2A(x) - p\big)_+.$$

*Here $r_+ = \max\{r, 0\}$.*

*Proof.* For $d(x) > 0$, Cauchy–Schwarz with respect to the finite measure $W(x, y) \, \mathrm{d}\mu(y)$ gives

$$A(x)^2 = \left( \int W(x, y) d(y) \, \mathrm{d}\mu(y) \right)^2 \le d(x) \int W(x, y) d(y)^2 \, \mathrm{d}\mu(y) = d(x) C(x).$$

If $d(x) = 0$, then $W(x, y) = 0$ for almost every $y$, and hence $A(x) = C(x) = 0$.

For the triangle estimate, apply $rs \ge r + s - 1$ to $W(x, y), W(x, z)$, multiply by $W(y, z) \ge 0$, and integrate in $y, z$. This gives $\tau(x) \ge 2A(x) - p$. Also $\tau(x) \ge 0$. □

## 3 The sharp scalar supporting inequality

The next lemma is the new analytic input. Its term $a - d^2$ is tailored to the last identity in Lemma 2.1.

**Lemma 3.1** (Dense rooted supporting plane)**.** *Let $p \in [1/2, 1]$, and let $d, a \in [0, 1]$ satisfy*

$$0 \le a \le d, \qquad a \ge d - (1 - p).$$

*Define the left side to be zero when $d = 0$. Then*

$$\frac{a^2}{d}(2a - p)_+ \ge 2p^4(1 - 4p) + p^3(10p - 3)d + 2p^2(3p - 1)(a - d^2). \tag{3}$$

*Proof.* Write the right side of (3) as

$$L_p(d, a) = \alpha + \beta d + \gamma(a - d^2),$$

where

$$\alpha = 2p^4(1 - 4p), \qquad \beta = p^3(10p - 3), \qquad \gamma = 2p^2(3p - 1) > 0.$$

If $d = 0$, then $a = 0$, while $L_p(0, 0) = \alpha \le 0$. We may therefore assume $d > 0$.

First suppose $a \le p/2$. The left side of (3) is zero, and $L_p(d, a)$ is increasing in $a$. If $d \le p/2$, it is enough to put $a = d$. Direct expansion gives

$$L_p(d, d) = -p^2 Q_1(d),$$

where

$$Q_1(d) = 2(3p - 1)d^2 + (-10p^2 - 3p + 2)d + 8p^3 - 2p^2.$$

The derivative of $Q_1$ at $d = p/2$ is $-4p^2 - 5p + 2 < 0$. Since $Q_1$ is convex, it is decreasing on $[0, p/2]$. Hence

$$Q_1(d) \ge Q_1(p/2) = \frac{p}{2}(9p^2 - 8p + 2) > 0;$$

the last quadratic has negative discriminant.

If instead $d \ge p/2$, it is enough to put $a = p/2$. Feasibility also forces $d \le 1 - p/2$. Here

$$L_p(d, p/2) = -p^2 Q_2(d),$$

where

$$Q_2(d) = 2(3p - 1)d^2 + (-10p^2 + 3p)d + 8p^3 - 5p^2 + p.$$

Its leading coefficient is positive and its discriminant is

$$-p(2p - 1)^2(23p - 8) \le 0.$$

Thus $Q_2(d) \ge 0$. This proves (3) when $a \le p/2$.

Now suppose $a \ge p/2$. Multiplication by $d > 0$ reduces the assertion to

$$G_p(d, a) := 2a^3 - pa^2 - dL_p(d, a) \ge 0. \tag{4}$$

For fixed $d$,

$$\frac{\partial^2 G_p}{\partial a^2} = 12a - 2p \ge 4p > 0.$$

Consequently the minimum on the feasible interval

$$\max\{p/2, d-(1-p)\} \le a \le d$$

occurs at an endpoint or at the unique stationary point. We check all possibilities exactly.

At $a = p/2$, expansion gives

$$G_p(d, p/2) = dp^2 Q_2(d) \ge 0.$$

At $a = d$, write $G_p(d,d) = dP_p(d)$, where

$$P_p(d) = 2(3p^3 - p^2 + 1)d^2 + (-10p^4 - 3p^3 + 2p^2 - p)d + 8p^5 - 2p^4.$$

The leading coefficient is positive and

$$\mathrm{disc}_d(P_p) = -p^2(p-1)^2 E(p) \le 0.$$

Indeed, on writing $u = p - 1/2 \ge 0$,

$$E(p) = 92u^4 + 196u^3 + 135u^2 + 36u + 2 > 0.$$

Therefore $P_p(d) \ge 0$.

For the remaining lower endpoint put

$$C_p(d) = G_p\big(d, d-(1-p)\big).$$

It is needed only for $d \in [1-p/2, 1]$. Its leading coefficient as a cubic in $d$ is $2(3p^3 - p^2 + 1) > 0$. For $p < 1$, exact expansion gives

$$\mathrm{disc}_d(C_p) = -8p^4(p-1)^4 D(p) < 0,$$

where, for $u = p - 1/2 \ge 0$,

$$\begin{aligned} D(p) = {} & 736u^{10} + 2624u^9 + 4448u^8 + 6756u^7 + 5532u^6 + 776u^5 \\ & + \frac{2331}{2}u^4 + \frac{2161}{4}u^3 + \frac{1185}{8}u^2 + \frac{197}{8}u + \frac{51}{32} > 0. \end{aligned}$$

Thus $C_p$ has exactly one real zero. At the left endpoint,

$$C_p(1-p/2) = \frac{p^2(2-p)}{4}(29p^3 - 46p^2 + 24p - 4) > 0.$$

The cubic in the last parentheses has discriminant $-304$, positive leading coefficient, and value $1/8$ at $p = 1/2$; its unique real zero is therefore less than $1/2$. It follows that the unique zero of $C_p$ lies below $1 - p/2$, and $C_p(d) > 0$ throughout the required interval. When $p = 1$, the same conclusion follows directly from

$$C_1(d) = 6d(d-1)^2 \ge 0.$$

It remains to check an interior stationary point. Solving $\partial G_p/\partial a = 0$ gives

$$a_* = \frac{p}{6}(1+\rho), \qquad \rho = \sqrt{1 + 12d(3p-1)}.$$

If $a_*$ belongs to the present region, then $a_* \geq p/2$ and hence $\rho \geq 2$. A direct factorization gives

$$G_p(d, a_*) = \frac{p^2(\rho+1)(6p-1-\rho)^2}{864(3p-1)^2} R_p(\rho).$$

Writing $r = \rho - 2 \geq 0$ and $u = p - 1/2 \geq 0$, the remaining factor is

$$R_p(\rho) = r^3 + (9+12u)r^2 + (24+66u+48u^2)r + 14 + 50u + 48u^2 > 0.$$

All endpoint and stationary-point minima are nonnegative, completing the proof. □

**Remark 3.2** (Why the density hypothesis is essential)**.** *The coefficients in Lemma 3.1, the lower constraint $a \geq d-(1-p)$, and the sign certificates all use $p \geq 1/2$. The lemma is not an unconditional tree-domination statement. This is exactly what allows it to avoid the forest-domination obstruction recorded in* `FAILED_PROOF_ATTEMPTS.md`*.*

# 4 The positive graphon theorem

**Theorem 4.1** (Triangle with a two-leaf broom)**.** *For every graphon $W$ on an arbitrary probability space, of edge density $p \in [1/2, 1]$,*

$$\boxed{t(B, W) \geq p^4(2p-1) = \Phi_B(p).}$$

*Proof.* For $d > 0$ define

$$h_p(d, a) = \frac{a^2}{d}(2a-p)_+,$$

and set $h_p(0,0) = 0$. By (1) and Lemma 2.2,

$$t(B, W) \geq \int_\Omega h_p(d(x), A(x))\, \mathrm{d}\mu(x).$$

Lemma 2.1 supplies the pointwise hypotheses of Lemma 3.1. Integrating that scalar inequality and using

$$\int d = p, \qquad \int (A - d^2) = 0,$$

gives

$$\begin{aligned} t(B, W) &\geq 2p^4(1-4p) + p^3(10p-3)p \\ &= p^4(2p-1), \end{aligned}$$

which is (2). □

At $p = 1/2$ the target is zero and the proof above remains valid. At $p = 1$, one may alternatively note that $0 \leq W \leq 1$ and $\int W = 1$ imply $W = 1$ almost everywhere, so both sides equal one. No division by $2p-1$ occurs.

For every integer $k \geq 2$, the balanced complete $k$-partite graphon has $p = 1 - 1/k$ and realizes equality throughout the proof: $d = p$, $A = p^2$, $C = p^3$, and $\tau = p(2p-1)$ almost everywhere. Thus the bound has the expected chromatic extremizers.

## 5 Atlas consequence, hidden-assumption audit, and limitations

**Corollary 5.1** (Atlas 100)**.** *NetworkX Atlas graph* 100*, with graph6 code* `EsCW`*, is positive. It has order six, size six, chromatic number three, chromatic polynomial*

$$\chi_{F_{100}}(x) = x(x-1)^4(x-2),$$

*and every graphon of edge density* $p \in [1/2, 1]$ *satisfies*

$$t(F_{100}, W) \geq p^4(2p-1) = \Phi_{F_{100}}(p).$$

*Proof.* The NetworkX edge list is

$$01,\ 02,\ 03,\ 34,\ 35,\ 45.$$

Vertices $3, 4, 5$ form a triangle, rooted at 3. Vertex 0 is joined to that root, and $1, 2$ are precisely the two leaves adjacent to 0. Hence the graph is exactly $B$, and Theorem 4.1 applies. □

All graphon integrands used above are bounded, nonnegative, and measurable, so Tonelli–Fubini applies on an arbitrary probability space. Every pointwise statement is understood almost everywhere. The proof uses ordinary homomorphism densities, not injective copies. It assumes neither a finite-step representation, regularity, constant degree, nor a positive minimum degree. The set $\{d = 0\}$ is handled without division.

For formal verification, the reusable new item is Lemma 3.1. A convenient implementation can split on $d = 0$, $a \leq p/2$, and $a \geq p/2$, exactly as above. The graphon layer then needs only Cauchy–Schwarz under the measure $W(x, y)\,\mathrm{d}\mu(y)$, the already used rooted-triangle inequality, the two identities of Lemma 2.1, and integration of the scalar supporting plane. The graph-specific module awaiting inspection is `lean/Taeyoung/Examples/Graph100.lean`.

The theorem covers exactly a triangle with one adjacent star centre and two leaves at that centre. It does not prove arbitrary pendant-tree closure, a star with another number of leaves, several such brooms, or brooms rooted at different triangle vertices. The supporting plane is special to the two-leaf exponent and the interval $p \geq 1/2$; none of these extensions may be inferred without a new scalar certificate.

Every polynomial identity, discriminant, positive coefficient shift, the Atlas isomorphism, and the chromatic polynomial used in this document are independently checked over exact rational arithmetic. This audit is only a check of the displayed algebra; the arbitrary-graphon argument is the proof given above.

## 6 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/Broom.lean`, ending at `satisfiesLowerBound_100`. Theorem 4.1 is proved in full, on all of $p \in [1/2, 1]$. The departure is inside Lemma 3.1, in its active case $a \geq p/2$.

**The departure: no cubic discriminant.** This note treats $G_p(d, a) = 2a^3 - pa^2 - dL_p(d, a)$ as a cubic in $a$ for fixed $d$ and runs a case analysis on its root structure. The formalization does it in one chain. In the coordinates

$$\alpha = a - p^2, \qquad \delta = d - p$$

— the point where the extremal graphon sits — the cleared inequality is the `ring` identity

$$a^2(2a-p) - d\,L_p(d,a) = p(6p-1)\alpha^2 - 2p^2(3p-1)\alpha\delta + p^3(8p-3)\delta^2 + 2\alpha^3 + 2p^2(3p-1)\delta^3. \quad (5)$$

Both cubic terms are absorbed using only the two facts that already hold in the active case — $a \geq p/2$ by assumption and $d \geq a \geq p/2$:

$$2\alpha^3 \geq (p-2p^2)\alpha^2, \qquad 2p^2(3p-1)\delta^3 \geq -p^3(3p-1)\delta^2,$$

each being $2\alpha^2(a-\frac{p}{2}) \geq 0$ and $2p^2(3p-1)\delta^2(d-\frac{p}{2}) \geq 0$ rearranged. What remains,

$$4p^2\alpha^2 - 2p^2(3p-1)\alpha\delta + p^3(5p-2)\delta^2,$$

is positive definite on $p \geq 1/2$, certified by

$$16p^2\big(4p^2\alpha^2 - 2p^2(3p-1)\alpha\delta + p^3(5p-2)\delta^2\big) = \big(8p^2\alpha - 2p^2(3p-1)\delta\big)^2 + 4p^4(11p^2-2p-1)\delta^2, \quad (6)$$

with $11p^2 - 2p - 1 \geq 3/4$ there. So the active case is four `ring` identities and three sign checks, and no cubic is analysed.

The inactive case $a \leq p/2$ follows this note exactly, including its two subcases $d \leq p/2$ (push $a$ up to $d$, then $Q_1 \geq Q_1(p/2) > 0$) and $d \geq p/2$ (push $a$ up to $p/2$, then $Q_2 \geq 0$ by its nonpositive discriminant $-p(2p-1)^2(23p-8)$).

**Everything else transcribes directly.** Lemma 2.1 is `Broom.le_pathOp` together with `K4Tail.pathOp_le_degree` and `BookTail.integral_pathOp`; Lemma 2.2 is `Broom.sq_pathOp_le` (the project's weighted Cauchy–Schwarz at weight $W(x,\cdot)$) and the pre-existing `Link.rootedTriangle_ge`. The division by $d$ is never performed: `Broom.mul_plane_le` states the pointwise bound multiplied by $d$, so the zero-degree set is an ordinary case rather than a convention. The identity (1) is `Broom.homDensity_broom`, through the generic peeling ladder of `Methods/Peeling.lean`.